estimate costs. And while many of these models do provide accurate pre-trade estimates, their lack of transparency does not allow portfolio managers to perform "what-if" analysis under various scenarios, or incorporate their own market expectations and alpha views into the process.

- Pre-trade cost calculations are often performed on the vendor's server. Managers need to pass their portfolio from their site to the vendor's server to obtain cost estimates. When portfolio holdings, data, or information leaves the manager's site there is always the potential for information leakage allowing the outside party to reverse engineer the manager's decision process. This could be detrimental to the fund's competitive edge.
- Pre-trade impact models incorporate the vendor's market expectations. These systems do not easily allow managers to revise factor expectations. For example, these models do not allow managers to change volatility, average daily volume, or expected liquidity over the trading horizon. If managers have better forecasts of explanatory variables there is no easy way for them to incorporate these values into the pre-trade estimates. And even if vendors make necessary provisions, there is still no way to do so without alerting these vendors of their own proprietary forecasts.
- Portfolio managers are very sensitive to alpha erosion. In other words, how much of their alpha will they capture given trading costs. But since managers are reluctant to pass these alpha estimates to any vendor's system these models are not able to structure strategies to minimize alpha erosion. Furthermore, managers are suspicious of any party providing alpha estimates for free and to a large array of customers.
- Constructing in-house market impact models is resource intensive and time consuming. Firms developing in-house models using their own trade data have the advantage of knowing the full order size, including shares cancelled and the decision price, and they can also incorporate their own proprietary market views into the cost estimate. This allows the market impact model to be customized for the fund's specific investment behavior. But this still does not allow the fund to perform thorough sensitivity analysis for a situation where they want to analyze an order that may be traded differently than they have in the past because they do not have any historical observations. These models could potentially suffer from in-sample bias.

The current approach being used by managers to incorporate TCA estimates into the stock selection process is to utilize systems such as vendor/broker web-sites, APIs, etc. Managers are asked to send their portfolios or